



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal to SRI
ANGALAPARAMESWARIAMMAN AG.**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

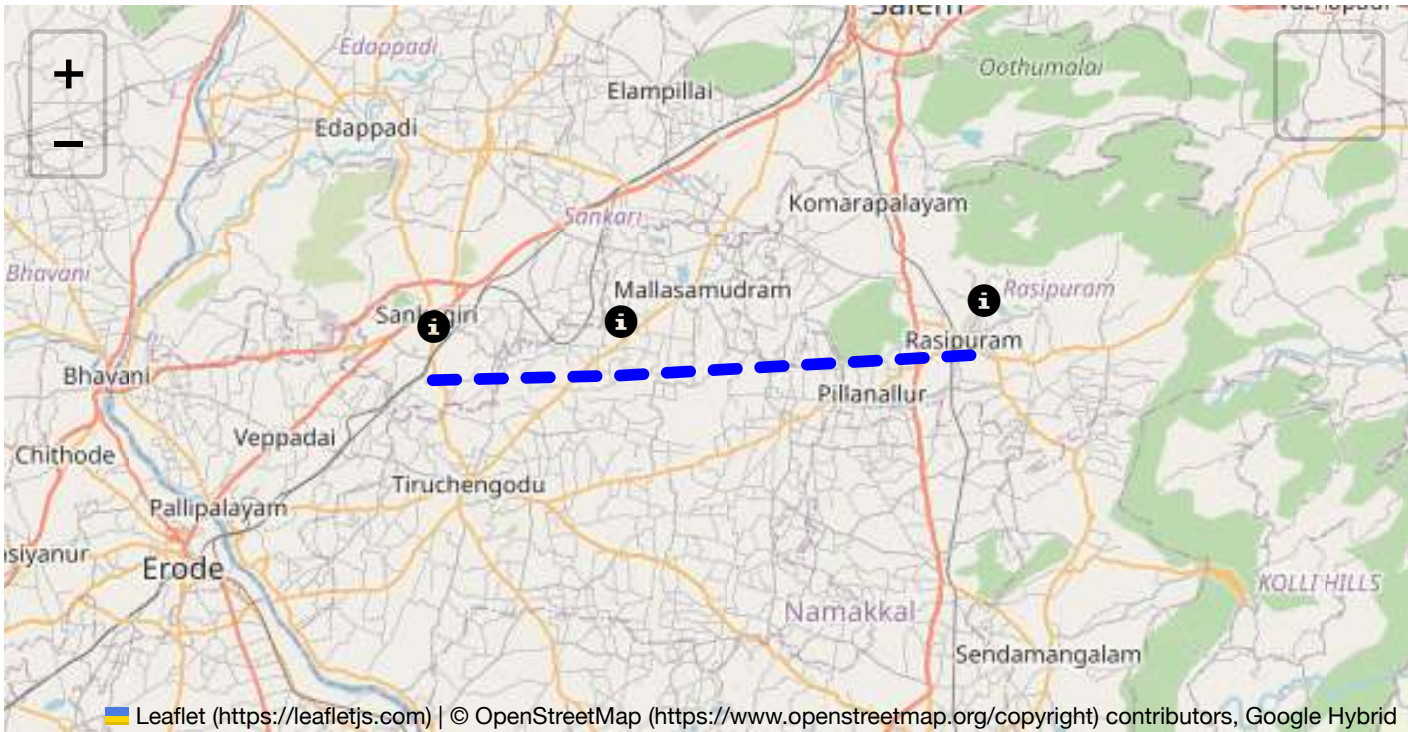
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

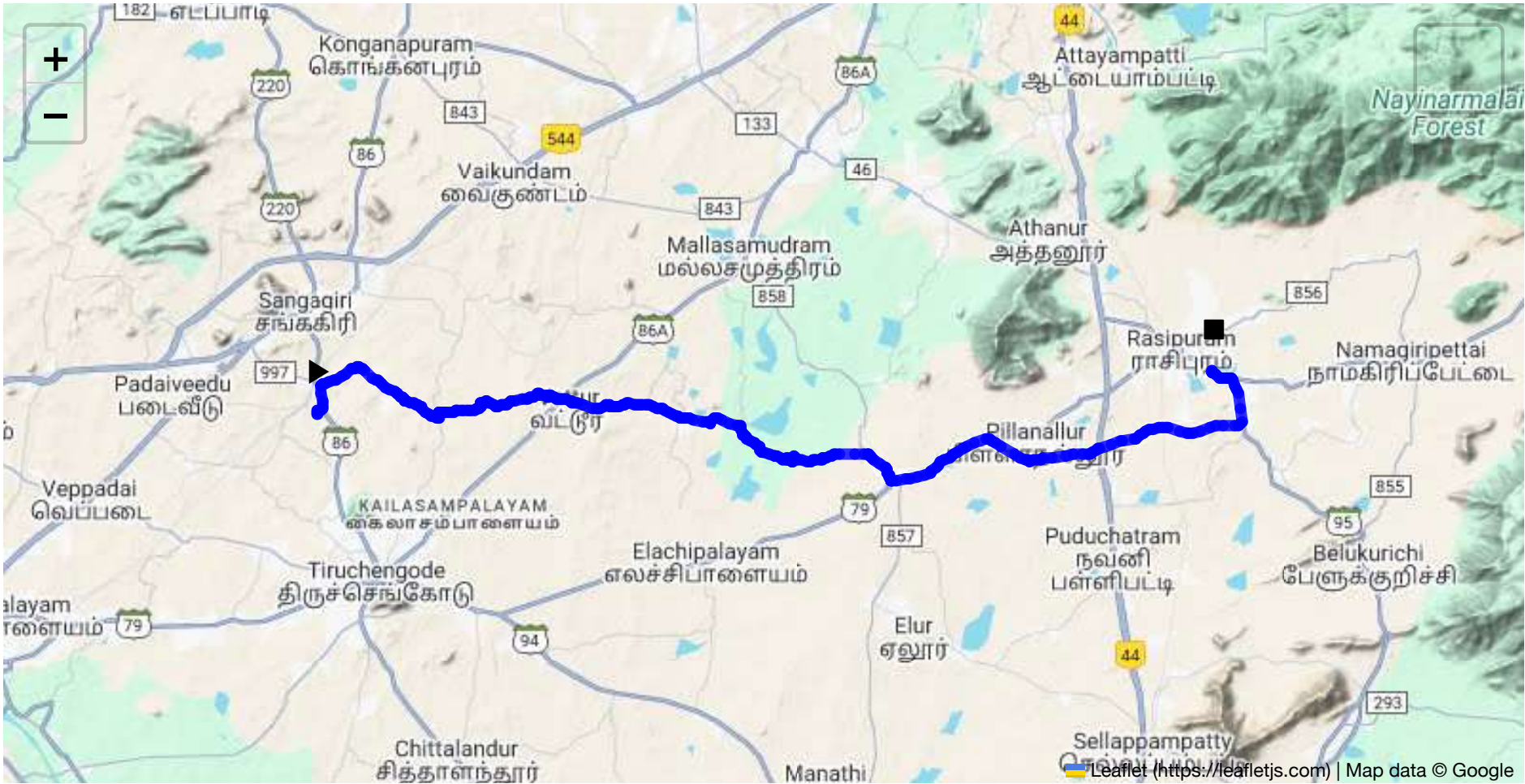


Route Summary:
Total Distance: 45.30 km
Estimated Duration: 1.1 hours

Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)

End: (11.453216, 78.196281)



Welcome to the Journey Risk Management Study

Below is a detailed analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to F53W+C5C, Rasipuram, Tamil Nadu 637408, India via Paruthipalli-Morur Rd, Vellayankadu, and Pillanatham, Tamil Nadu 637503, India. The information considers the requirements for a truck driver transporting hazardous materials.

- Overview of the Route Map:**
 - The route covers approximately 45.30 kilometers. Predominantly it passes through rural and semi-urban areas characterized by moderately trafficked roads. The route provides a mixture of state highways and local roads.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
 - The region experiences a tropical climate with hot summers, a heavy monsoon season, and mild winters.
 - Monsoon (June to September) can cause waterlogging and slippery road conditions. Fog during the early winters (December to January) might affect visibility.
- Traffic Patterns and Congestion:**
 - Peak traffic hours are typically in the morning between 7:30 – 9:30 AM and evening from 5:00 – 7:00 PM.
 - Sangagiri junctions and entry/exit points near the small towns along the route may experience congestion, especially on market days and local festivals.
- Assessment of Road Quality and Infrastructure:**
 - The major roads are in relatively good condition, but some rural stretches may have potholes and uneven surfaces.
 - Watch for narrow bridges and roads in some villages.

- Roadwork can occasionally cause minor delays.
5. **Suggestions for Alternative Routes for Emergencies:**
- Depending on the location of the emergency, options include reverting to the Salem or Namakkal highways which are broader and have better-maintained infrastructure.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:**
- Transporting hazardous materials requires adhering to Tamil Nadu Transport Department rules, which includes permits specifying routes and adherence to load restrictions.
 - Driving restrictions may apply during certain hours or local holidays.
7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**
- The region has reported some incidents involving tankers, mostly due to human error or mechanical failures.
 - Awareness campaigns on safe transport practices have been increasing.
8. **Environmental Considerations and Sensitive Areas:**
- Be mindful of the farmers and rural communities relying on these routes; ensure waste or hazardous spillage does not occur.
 - Consideration for routes passing close to schools should be given priority to ensure safety.
9. **Analysis of Communication Coverage:**
- The route generally benefits from good communication coverage, but expect potential dead spots near remote stretches or when passing through dense foliage.
10. **Estimated Emergency Response Times:**
- Emergency services in urban clusters like Sangagiri typically can respond within 20-30 minutes.
 - Response could take longer (up to 60 minutes) in more rural and remote areas.
11. **Overall Summary of Risk Assessment:**
- **Moderate Risk:** While the road conditions remain fairly stable, certain stretches are vulnerable to weather conditions and increased traffic during peak times.
 - Hazardous materials require adherence to strict route and handling guidelines.
 - Alternative routes should be planned considering congestion and potential environmental impacts.
 - Effective communication, continuous weather monitoring, and pre-defined action plans can significantly mitigate risks.

In conclusion, while the route is generally straightforward, careful planning and preparation regarding traffic, weather, and specific driving regulations for hazardous materials are essential for safe transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45478, 77.88925	30 KM/Hr
5	Turn	High	11.43675, 77.91693	15 KM/Hr
6	Turn	High	11.43684, 77.91706	15 KM/Hr
7	Turn	High	11.43926, 77.91705	15 KM/Hr
8	Turn	Medium	11.43930, 77.92282	30 KM/Hr
9	Turn	Medium	11.43967, 77.92345	30 KM/Hr
10	Turn	Medium	11.43979, 77.93009	30 KM/Hr
11	Turn	Medium	11.44030, 77.93066	30 KM/Hr
12	Turn	Medium	11.44067, 77.93203	30 KM/Hr
13	Turn	High	11.44088, 77.93223	15 KM/Hr
14	Turn	Medium	11.44127, 77.93218	30 KM/Hr
15	Turn	Medium	11.44163, 77.93244	30 KM/Hr
16	Turn	High	11.44243, 77.93244	15 KM/Hr
17	Turn	Medium	11.44285, 77.93440	30 KM/Hr
18	Turn	Medium	11.44093, 77.93821	30 KM/Hr
19	Turn	Medium	11.44550, 77.95329	30 KM/Hr
20	Turn	Medium	11.44369, 77.96036	30 KM/Hr
21	Turn	Medium	11.44395, 77.96298	30 KM/Hr
22	Turn	Medium	11.44307, 77.96476	30 KM/Hr
23	Turn	High	11.44236, 77.96508	15 KM/Hr
24	Turn	Medium	11.44104, 77.98026	30 KM/Hr
25	Turn	Medium	11.44210, 77.98526	30 KM/Hr
26	Turn	Medium	11.44012, 77.99840	30 KM/Hr
27	Turn	Medium	11.43522, 78.01519	30 KM/Hr
28	Turn	Medium	11.43719, 78.01784	30 KM/Hr
29	Turn	High	11.43441, 78.02649	15 KM/Hr
30	Turn	Medium	11.43116, 78.02700	30 KM/Hr
31	Turn	Medium	11.42462, 78.03450	30 KM/Hr
32	Turn	Medium	11.42324, 78.03833	30 KM/Hr
33	Turn	High	11.42328, 78.03858	15 KM/Hr
34	Turn	High	11.42366, 78.03881	15 KM/Hr
35	Turn	Medium	11.42215, 78.04127	30 KM/Hr
36	Turn	Medium	11.42213, 78.04275	30 KM/Hr
37	Turn	Medium	11.42224, 78.04290	30 KM/Hr
38	Turn	Medium	11.42256, 78.04302	30 KM/Hr
39	Turn	High	11.42263, 78.04314	15 KM/Hr
40	Turn	High	11.42150, 78.04467	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
41	Turn	Medium	11.42173, 78.04489	30 KM/Hr
42	Turn	High	11.42312, 78.04534	15 KM/Hr
43	Turn	High	11.42318, 78.04547	15 KM/Hr
44	Turn	Medium	11.42120, 78.05015	30 KM/Hr
45	Turn	High	11.42195, 78.05572	15 KM/Hr
46	Turn	High	11.42259, 78.05580	15 KM/Hr
47	Turn	Medium	11.42326, 78.05838	30 KM/Hr
48	Turn	Medium	11.42445, 78.06044	30 KM/Hr
49	Turn	Medium	11.42415, 78.07233	30 KM/Hr
50	Turn	Medium	11.41950, 78.07852	30 KM/Hr
51	Turn	Medium	11.41791, 78.07940	30 KM/Hr
52	Blind Spot	Blind Spot	11.41453, 78.08062	10 KM/Hr
53	Turn	Medium	11.41466, 78.08070	30 KM/Hr
54	Turn	High	11.42962, 78.11578	15 KM/Hr
55	Turn	Medium	11.42450, 78.15278	30 KM/Hr
56	Turn	High	11.43542, 78.20754	15 KM/Hr
57	Turn	Medium	11.45096, 78.20358	30 KM/Hr
58	Turn	High	11.45346, 78.19637	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
5	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Sri Vinayaga Matriculation Higher Secondary School	11.4233682, 78.0736517	30 km/h	Medium
1	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
2	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
4	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43675, 77.91693



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43684, 77.91706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43930, 77.92282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44093, 77.93821



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44395, 77.96298



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44307, 77.96476



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44236, 77.96508



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44104, 77.98026



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44210, 77.98526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44012, 77.99840



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43522, 78.01519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43719, 78.01784



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43441, 78.02649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43116, 78.02700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42462, 78.03450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42324, 78.03833



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42328, 78.03858



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42366, 78.03881



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42215, 78.04127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42213, 78.04275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42224, 78.04290



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42256, 78.04302



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42263, 78.04314



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42150, 78.04467



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42173, 78.04489



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42312, 78.04534



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42318, 78.04547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42120, 78.05015



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42195, 78.05572



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42259, 78.05580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42326, 78.05838



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42445, 78.06044



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42415, 78.07233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41950, 78.07852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41791, 78.07940



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41466, 78.08070



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42962, 78.11578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45096, 78.20358



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45346, 78.19637
